Reviewed Preprint
v1 • September 9, 2025
Not revised

Reviewed Preprint
v2 • February 12, 2026
Revised by authors

Tolerance to Lung Infection in TWIK2 K$^+$ Efflux Mediated Macrophage Trained Immunity

Josh Thompson[1,2], Yufan Li[3], Yuanling Song[3], Anas Anas[1,2], Jaewon Cho[1,2], Ki-Wook Kim[1,2], Asrar B Malik[1,2], Jingsong Xu[1,2,4]

[1]Department of Pharmacology and Regenerative Medicine, University of Illinois, Chicago, United States • [2]Center for Lung and Vascular Biology, University of Illinois, Chicago, United States • [3]Shanghai Key Laboratory of Lung Inflammation and Injury, Department of Pulmonary Medicine, Zhongshan Hospital, Fudan University, Shanghai, China • [4]Department of Hematology, Children's Hospital of Chongqing Medical University, Chongqing, China

For correspondence:
asrarb.malik@gmail.com
jingsong.xu@hotmail.com

Competing interests: No competing interests declared

Funding: See page 16

Reviewing editor: Florent Ginhoux, Singapore Immunology Network, Singapore

eLife Assessment

This study presents **valuable** data suggesting that ATP-induced modulation of alveolar macrophage (AM) functions is associated with NLRP3 inflammasome activation and enhanced phagocytic capacity. While the in vivo and in vitro data reveal an interesting phenotype, the evidence provided is **incomplete** and does not fully support the paper's conclusions. Additional investigations would be of value in complementing the data and strengthening the interpretation of the results. This study should be of interest to immunologists and the mucosal immunity community.

https://doi.org/10.7554/eLife.107536.2.sa3

Abstract

Lung macrophages such as alveolar macrophages (AMϕ) are essential for innate immune function in the lungs. It is now apparent that macrophages can be trained to become better at attacking infections. Although trained immunity is thought to result from metabolic and epigenetic reprogramming, the underlying mechanisms remain unclear. Here, we report that macrophages can be trained by extracellular ATP, which is ubiquitously released during inflammation. ATP ligates the canonical Purinergic Receptor 2 subtype X7 receptor (P2X7) to mediate endosomal Two-pore domain Weak Inwardly rectifying K$^+$ channel 2 (TWIK2) translocation into the plasma membrane (PM). This endows the cells to transit to a 'ready' state for microbial killing in two directions: first, K$^+$ efflux via PM-TWIK2 induces NLRP3 inflammasome activation, which further activates metabolic pathways; second, upon bacterial phagocytosis, PM-TWIK2 internalizes into phagosome membrane with proper topological orientation, where TWIK2 mediates K$^+$ influx into phagosomes to control pH and ionic strength favoring bacterial killing. Therefore, the enhanced association of TWIK2 in phagosomal and plasma membranes signaled by danger-associated molecular patterns (DAMPs), such as ATP, mediates trained immunity in macrophages and enhances the microbiocidal activity.

Introduction

Immune responses are classically divided into innate immune responses, which react rapidly and nonspecifically upon encountering a pathogen, and adaptive immune responses, which are slower to develop show a measured response that is specific and the result of acquiring immunological memory (Netea et al., 2016). Although innate immunity is considered a non-specific, short-lived phenomenon as opposed to adaptive immunity, which is long-lived and highly specific, recent studies have shown that innate immunity can display adaptive immunity characteristics after challenge with pathogens or their products (Netea et al., 2016). Trained immunity mediates protection against heterologous infections and is thought to be due to epigenetic and functional reprogramming of innate immune cells (Netea and Joosten, 2018). However, trained immunity also exhibits maladaptive effects in chronic inflammatory conditions such as cardiovascular diseases and autoinflammatory syndromes (Netea and Joosten, 2018).

Lung resident macrophages such as AMϕ are key effector cells of the innate immune response in the lungs, which not only play a key role in killing bacterial pathogens through processes such as phagocytosis or secretion of antimicrobial peptides, but also play an important role in the restoration of homeostasis (Allard et al., 2018). The efficiency of bacterial killing is an essential determinant of the ability to resolve lung inflammation and injury as seen in patients with acute lung injury (ALI) and acute respiratory distress syndrome (ARDS) (Sevransky et al., 2009). Being located at the interface of the airways and the environment, lung AMϕ exhibit a distinctive cellular profile that tightly regulates their activation state to avoid excessive inflammation (Bain and MacDonald, 2022). Due to their unique properties, lung AMϕ are important candidates for understanding trained immunity. Our current knowledge about innate immune memory responses of lung macrophages and the underlying mechanisms remains limited.

NLRP3 inflammasome is a key determinant of acute immune responses as seen in ALI/ARSD (Grailer et al., 2014; Sefik et al., 2022; Swanson et al., 2019). Activation of NLRP3 inflammasome complex is a multi-step process involving the assembly of key proteins and activation of caspase-1 which cleaves pro-Interleukin-1β (pro-IL-1β) to release the active form of this inflammatory cytokine (He et al., 2016; Swanson et al., 2019). However, little is known about the triggers that initiate the activation of NLRP3 complex. An essential mechanism of NLRP3 assembly is the efflux of K^+ at PM (Franchi et al., 2007; Munoz-Planillo et al., 2013; Petrilli et al., 2007) through the K^+ channel TWIK2 (Di et al., 2018; Enyedi and Czirjak, 2010). The efflux of K^+ generates regions of low intracellular K^+, which promote a conformational change of inactive NLRP3 to facilitate NLRP3 assembly and activation (Tapia-Abellan et al., 2021). TWIK2 mediated K^+ efflux thus serves as a checkpoint for the initiation of trained immunity as well as maladaptive inflammatory signaling mediated by NLRP3 (Di et al., 2018). This function of TWIK2 in initiating macrophage trained immunity remains uncertain.

Our previous work outlined a mechanism wherein extracellular ATP induces Rab-11 mediated outward translocation of the TWIK2 K^+ efflux channel (Huang et al., 2023). Membrane-associated TWIK2 was sufficient to drive NLRP3-activating potassium efflux, significantly enhancing inflammatory innate immune function (Di et al., 2018; Huang et al., 2023). Here, we show that TWIK2 after PM insertion (induced by ATP) is re-internalized into macrophages upon infection while remaining associated with the phagosome membrane. TWIK2 is localized in the phagosome membrane and increases phagosome ionic strength by transporting K^+ from the cytosol into the phagosome, which favors phagosomal protease activation thus enhancing bactericidal activity. Therefore, TWIK2 functions both at the PM and in phagosomes to optimize bacterial killing and the channel can be modulated to promote macrophage training. The regulatable association of TWIK2 with the PM and phagosomes triggered by ATP mediates trained immunity in macrophages yet avoids excessive inflammation.

Results

TWIK2 is required for ATP induced macrophage training

Lung AMϕ are exposed to inhaled particles and microbes and thus represent ideal cells to study trained immunity. Since ATP is widely appreciated as a ubiquitous DAMP (Coutinho-Silva and Savio, 2021) that can activate NLRP3 via P2X7-TWIK2 signaling in macrophages (Di et al., 2018; Huang et al., 2023), we investigated the role of ATP in induction of immune memory. WT, *Twik2*$^{-/-}$, *P2x7*$^{-/-}$, and *Nlrp3*$^{-/-}$ mice were intranasally exposed to ATP (1mM, i.n.) and AMϕ were isolated on day 7 to determine acquisition of training. The cells were further challenged with GFP-expressing *Pseudomonas aeruginosa* (GFP-*PA*, Fig. 1A). To assess the microbial killing activity, we measured the relative survival of phagocytosed *PA* within AMϕ in the presence or absence of extracellular ATP, as well as the residual bacterial load. AMϕ from ATP treated WT mice showed markedly augmented microbial killing as compared with controls (Fig. 1B). Similar results were observed in bone marrow-derived macrophages (BMDM) trained with ATP *in vitro* (Fig. 1C). Importantly, bactericidal activity remained unaffected in AMϕ or BMDM from TWIK2, P2X7 or NLRP3 null mice (Fig. 1B), indicating a crucial role of P2X7-TWIK2-NLRP3 signaling in ATP-mediated AMϕ training. These findings support a model that ATP functions through TWIK2 channel to activate training. Indeed, ATP-trained mice had a better survival rate after being infected intratracheally with *PA*, as compared with untrained controls (Fig. 1D). The induction of memory was not observed in AMϕ treated with other DAMPs such as histone (150 μg/ml) (Jung et al., 2022) or NAD (0.5 mM, Fig. 1E) (Adriouch et al., 2012) highlighting the importance of ATP signaling.

The ATP-trained AMϕ also produced greater amount of pro-inflammatory IL-1β as compared to control AMϕ after PA (Fig. 1F), and this increment was not observed in TWIK2 null AMϕ (Fig. 1F). Other cytokines such as IL-6 and IL-18 were not significantly changed before and after ATP training (Fig. 1G, H). It is worth noting that the populations of immune cells such as neutrophils, monocytes, AMϕ and eosinophils were not significantly altered after ATP treatment in lungs (Fig. 2A-D), indicating that ATP-induced training may not require newly recruited immune cells into the lungs.

ATP induces endosomal TWIK2 PM translocation in macrophages

We hypothesized that TWIK2 translocation to plasma membrane (PM) and phagosome determines its role in macrophage training, therefore, we studied TWIK2 dynamics by imaging TWIK2 PM translocation using TWIK2-GFP. We observed in RAW264.7 cells that TWIK2 translocated to the PM within 2 min post-ATP (Fig. 3A); this observation was confirmed by membrane fractioned TWIK2 (Fig. 3B). After ATP training, fluorescence recovery after photobleaching (FRAP) demonstrated a significantly reduced mobile fraction for membrane bound TWIK2, indicating that the diffusion of membrane-translocated TWIK2 is indeed constrained by the membrane (Fig. 3C). Surprisingly, PM TWIK2 could be positioned at the PM for as long as 6 days following ATP activation (Fig. 3D). These data suggest PM association may sustain functionally relevant macrphage training. Interestingly, LPS itself showed no such effect on TWIK2 PM association (Fig. 3D) suggesting the importance of ATP signaling in activating the process.

Upon phagocytosis, TWIK2 translocation to the phagosomal membrane may promote enhanced bactericidal activity through regulating phagosome acidification and ionic strength (Reeves et al., 2002). Therefore, we compared TWIK2 localization during phagocytosis in TWIK2-GFP expressing RAW264.7 cells trained with ATP vs. control cells. While both groups (ATP trained and untrained) efficiently endocytosed zymosan, ATP training markedly increased TWIK2-GFP phagosomal membrane translocation as compared to untrained control cells (Fig. 4A, B). Thus, TWIK2 after its PM insertion is internalized in its proper topology in the membrane of particle-laden phagosomes.

Next, we compared phagosomal K^+ changes after bone marrow-derived macrophages (BMDM) phagocytosed zymosan. We found that untrained cells exhibited little to no net phagosome K^+ influx whereas ATP trained BMDM markedly increased net phagosome K^+ influx relative to the cytoplasmic concentration (indicated by a specific K^+ dye, ION K^+, green) (Fig. 4C, D). Thus, TWIK2 translocates to phagosome membrane upon phagocytosis of trained macrophages and increases the phagosome ionic strength, which favors bacterial clearance. K^+ influx in phagosomes via TWIK2 renders phagosomes hypertonic, which is known to activate hydrolytic proteases through the fusion of granules(Vieira et al., 2002). Indeed, ATP-induced macrophage training effects, as measured by augmented bactericidal activity, were diminished in macrophages treated with protease inhibitors (Fig. 5A).

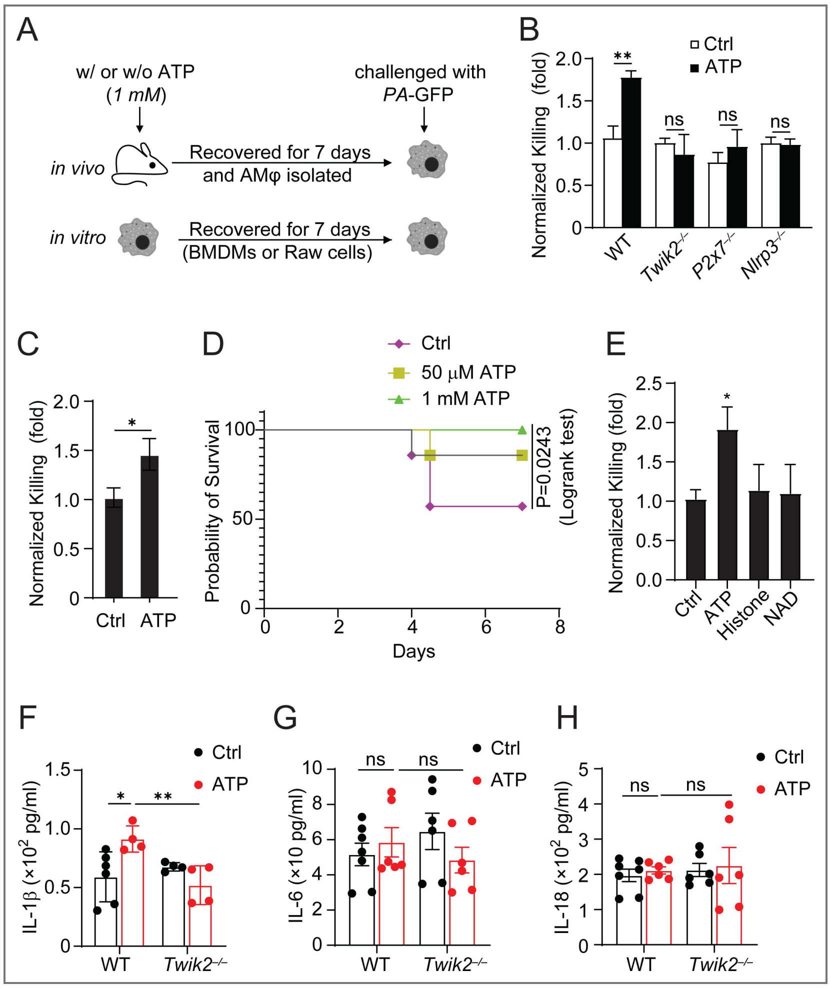


Figure 1. **Central role of TWIK2 in triggering ATP-induced macrophage training of bacterial killing.**

A. Diagram of the ATP-induced AMφ training model in vivo. WT or KO mice were exposed to ATP (1mM, i.n.) and AMφ were isolated on day 7 to determine acquisition of training. For in vitro training, BMDMs or RAW cells were trained with or without ATP (1mM) and recovered for 7 days. **B.** AMφ subjected to training as described in A were incubated with live PA-GFP (MOI:1), and assessed for relative bacterial load clearance. **C.** BMDMs subjected to in vitro training (1mM ATP for 7 days as described in A) were incubated with live PA-GFP (MOI:1) and were assayed for relative PA bacterial killing. **D.** Survival of control or ATP pretrained mice after receiving 40mL of 5×10^6 c.f.u. of *P. aeruginosa* intranasally instilled, n = 5. **E.** Comparison of bacterial load clearing activity of AMφ trained with various DAMPs: ATP, histone or NAD. ATP showed the most dramatic response. **F-H.** Changes in concentrations of IL1-β (F), IL-6 (G), and IL-6 (H) in ATP-trained AMφ after PA infection. ∗, p<0.05, ∗∗, p<0.01, data are from three independent experiments (mean and s.e.m.).

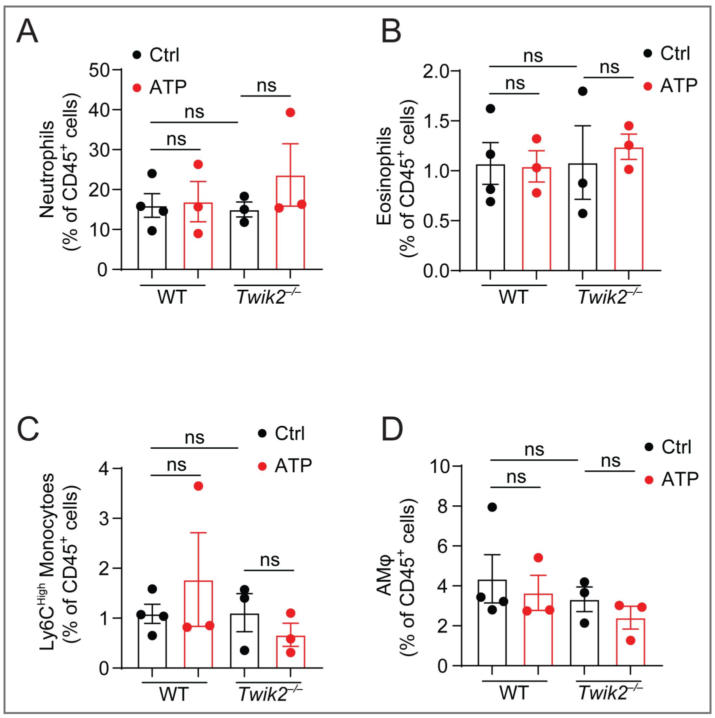


Figure 2. Immune cell recruitment in the lungs after ATP training.

A-D. ATP-induced recruitment of neutrophils (A), eosinophils (B), monocytes (C) and AMφ (D) in lung tissue. Data are from three independent experiments (mean and s.e.m.).

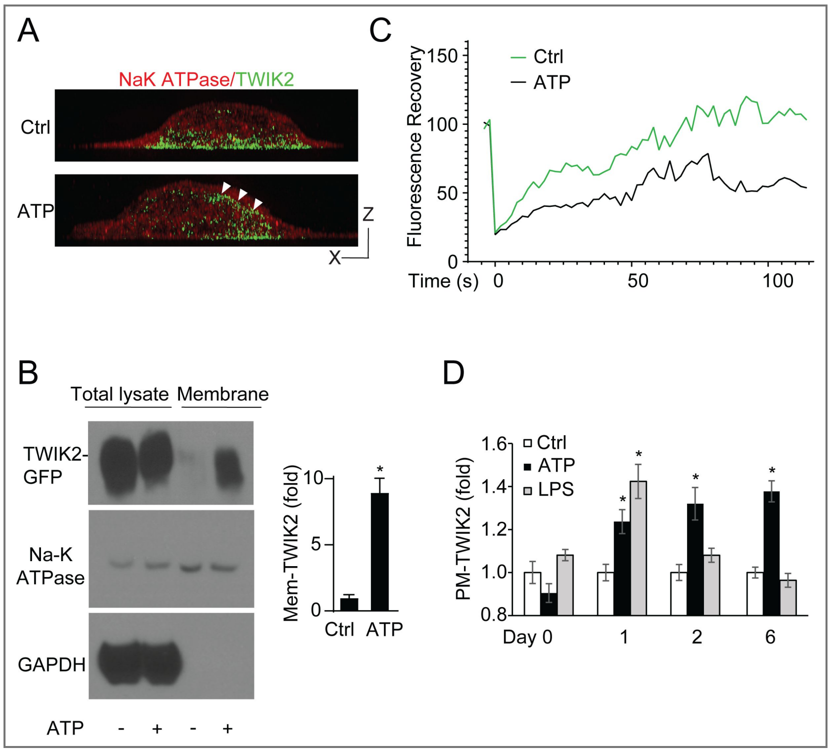


Figure 3. Assessment of TWIK2 PM and phagosome translocation.

A. Images of RAW cells expressing TWIK2-GFP post-ATP challenge (1mM, PM stained in red). White arrowheads show PM translocated TWIK2. Bar, 1μm. **B.** Western blot analysis of ATP trained or untrained macrophage lysate after membrane isolation. Membrane and non-membranous fractions were compared to assess the relative abundance of TWIK2 and the Na-K ATPase with and without ATP training. **C.** Fluorescence Recovery After Photobleach of membrane-associated TWIK2-GFP in ATP- or control-treated cells. **D**. Ratiometric fluorescence intensity analysis of TWIK2-GFP signal within 1 micrometer of the plasmalemma to the corresponding TWIK2-GFP cytoplasmic intensity, highlighting enhanced and long-term membrane association after ATP treatment. Duration of PM-TWIK2 untreated or ATP or LPS treated cells is shown. *, $P<0.05$, compared with control. Data are representative of (A, B-left panel) or from (B-right graph, D) three independent experiments, mean and s.e.m. in B, D.

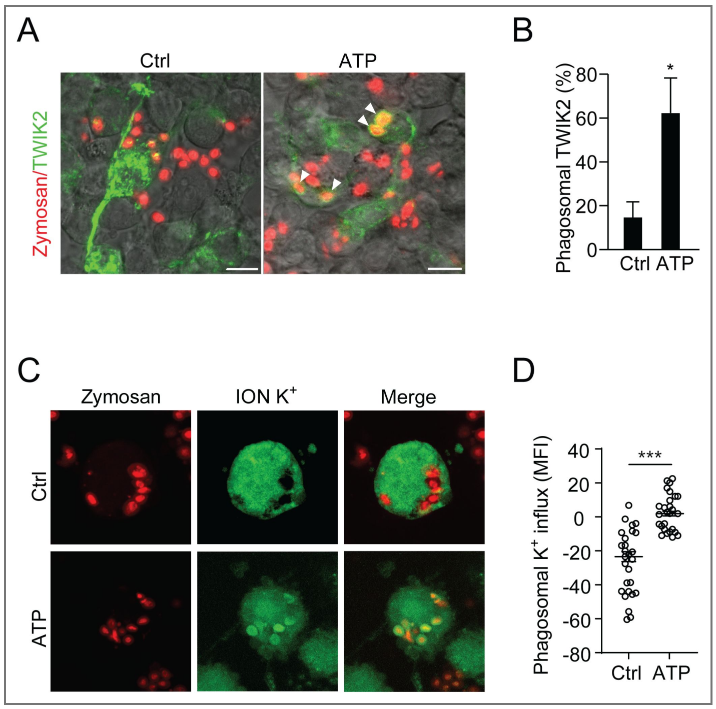


Figure 4. **PM-TWIK2 re-internalizes into phagosomes upon phagocytosis.**

A. Fluorescence images of TWIK2-GFP expressing RAW 264.7 cells phagocytosing AF594-conjugated zymosan (red) with or without ATP treatment. **B.** Quantification of TWIK2 localization frequency in A. We observed significantly more frequent phagosome-associated TWIK2 in ATP-trained cells vs. controls. Arrowheads point to phagosomes with TWIK2-GFP. ∗, $p<0.01$ compared with untreated control (n = 3). **C, D**, Representative images (C) and quantification (D) of K^+ enrichment (indicated by a specific K^+ dye, ION K^+, green) in phagosomes relative to cytoplasm of control and ATP-trained BMDMs, ∗∗∗, $p < 0.001$. Data are representative of (A, C) or from (B, D) three independent experiments, mean and s.e.m. in B, D.

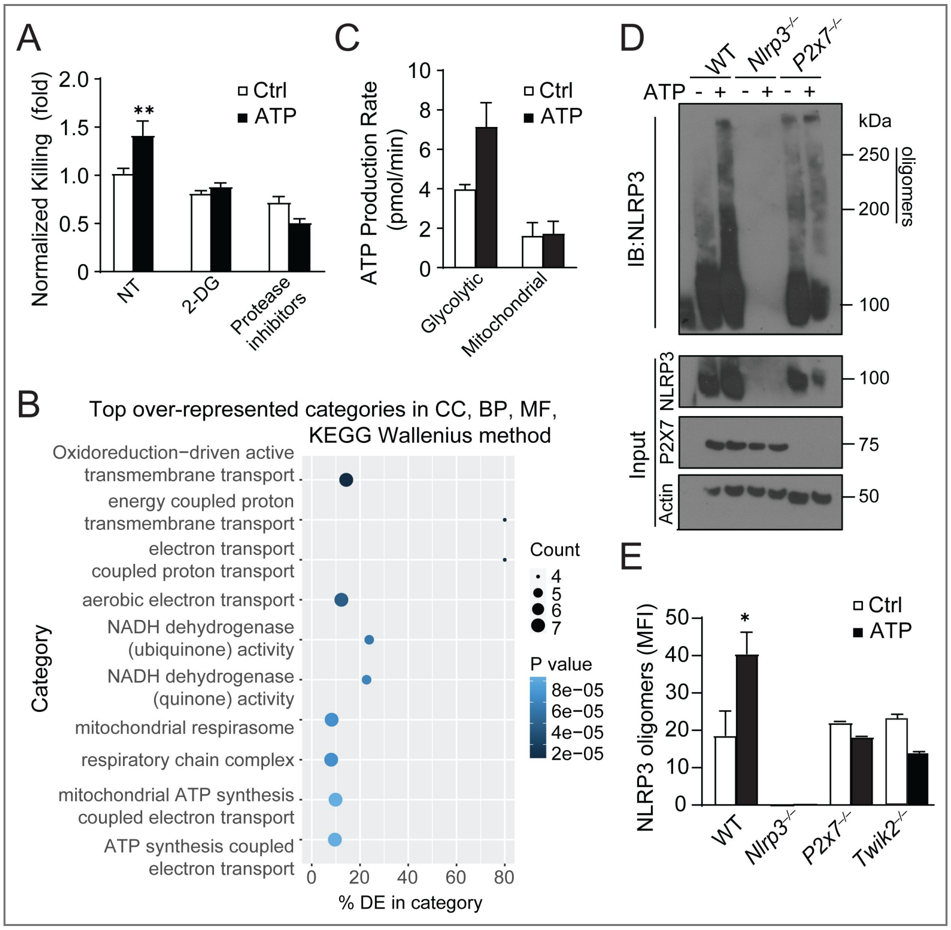


Figure 5. **ATP training regulates transcriptional profile and metabolic reprogramming of macrophages.**

A. Normalized *PA* killing ability of BMDMs trained with or without ATP in the presence of metabolic inhibitors 2DG or protease inhibitors. **B.** RNA isolated from BMDMs trained with or without prior ATP treatment was mapped to the *Mus musculus* genome and subsequently sorted into GO terms in Galaxy. Shown are the top 10 GO categories overrepresented in ATP trained BMDMs. **C.** Basal glycolytic and mitochondrial metabolism in BMDMs trained with or without ATP seeded equivalently 2 days prior to assessment as exhibited by Seahorse assay. **D, E.** Western blot (D) and quantification (E) of NLRP3 assembly in BMDMs trained with or without 1mM ATP and challenged with 1.0 MoI *PA* for 2 hours. These cells were lysed with DSP crosslinker and assessed for NLRP3 oligomerization. Data are representative of (B, D) or from (A, C, E) three independent experiments, mean and s.e.m. in A, C, E.

ATP-TWIK2 signaling regulates transcriptional profile and metabolic reprogramming of macrophages

Since epigenetic remodeling and metabolic reprogramming are two processes that may be fundamental to the mechanisms of trained immunity (Netea and Joosten, 2018), we further examined how ATP training regulates transcriptional profile in macrophages. To identify transcriptional changes of ATP-exposed and control cells, BMDMs were trained with or without ATP for 7 days and evaluated by RNA-seq. GO analysis of the bulk transcriptome was performed in these trained and control cells. We discovered that the differentially expressed functional groups most markedly upregulated after ATP training are metabolic genes (Fig. 5B). To further assess the effect of the differentially regulated metabolic programs of trained and untrained macrophages, we evaluated each set via the SEAHORSE ATP production rate assay. This revealed that ATP-treated BMDMs displayed increased metabolic activity, specifically by way of glycolytic ATP production (Fig. 5C). Importantly, ATP-induced augmented bactericidal activity was abolished in macrophages treated with competitive metabolic inhibitor 2-deoxy-d-glucose (2-DG) (Fig. 5A).

Due to the dependance of bacteria killing on NLRP3 inflammasome, we assessed the role of ATP training on the NLRP3 inflammasome activation. By lysing cells in the presence of cross-linker, we assessed the extent of high molecular weight NLRP3 oligomerization and complex formation in BMDMs with two hours of bacteria co-incubation. In untrained macrophages, exposure to bacteria resulted in the formation of 200 and 250 kDa complexes (Fig. 5D, E), indicating NLRP3 inflammasome assembly. This oligomerization was enhanced in ATP trained BMDMs, but was absent in P2X7 or TWIK2 knockout macrophages, in which NLRP3 oligomerization exhibited no difference between trained or nontrained P2X7, or TWIK2 null BMDMs (Fig 5D, E). NLRP3 is known to interact with some metabolic enzymes such as 6-phosphofructo-2-kinase/fructose-2,6-bisphosphatase 3 (PFKFB3) (Finucane et al., 2019) and phosphoglycerate kinase 1 (PGK1)(Zhu et al., 2024). Hence, ATP-mediated macrophage training may enhance metabolic activity through the NLRP3 inflammasome.

We also performed ATAC seq analysis in ATP-trained and control cells to evaluate regions of differentially accessible chromatin (Supp. Fig. 1). Interestingly, NLRP3 is among the genes with consistently more accessible chromatin within 2 kb of the upstream promoter region (Supp. Fig. 1), highlighting the importance of NLRP3 inflammasome in mediating ATP training.

ATP training rescues pneumonia-induced immunosuppression

Bacterial pneumonia after influenza is a leading cause of severe respiratory infections worldwide (Metzger and Sun, 2013). Unlike adenoviral infection (Yao et al., 2018), influenza infection abrogates macrophage-dependent bacteria clearance (Verma et al., 2020), indicating that context (type, dose and duration of infection) may be critical for macrophage phenotypes. We utilized a double-infection model to mimic this clinical scenario. Mice were first subjected to bacterial (*E. coli*) pneumonia, left to recover with or without ATP training (1 mM, i.t) for 7 days, and then infected i.t. with GFP-*PA* to cause secondary pneumonia (Fig. 6A). We found that the bacterial burden in lung tissue of mice doubly infected with both *E. coli* and *PA* was greater than in mice without the primary infection but that the extent of this bacterial load increase was markedly reduced with ATP treatment one week prior to secondary infection (Fig. 6B), which was not observed in other DAMP treatment such as NAD (Fig. 6B). This rescue effect by ATP was not observed in P2X7, NLRP3 or TWIK2 KO mice, indicate the importance of P2X7-TWIK2-NLRP3 singling in ATP mediated macrophage training (Fig. 6C). These results demonstrate that macrophages develop with impaired capacity to ingest or kill bacteria during pneumonia and the dysfunction can be rescued by ATP training.

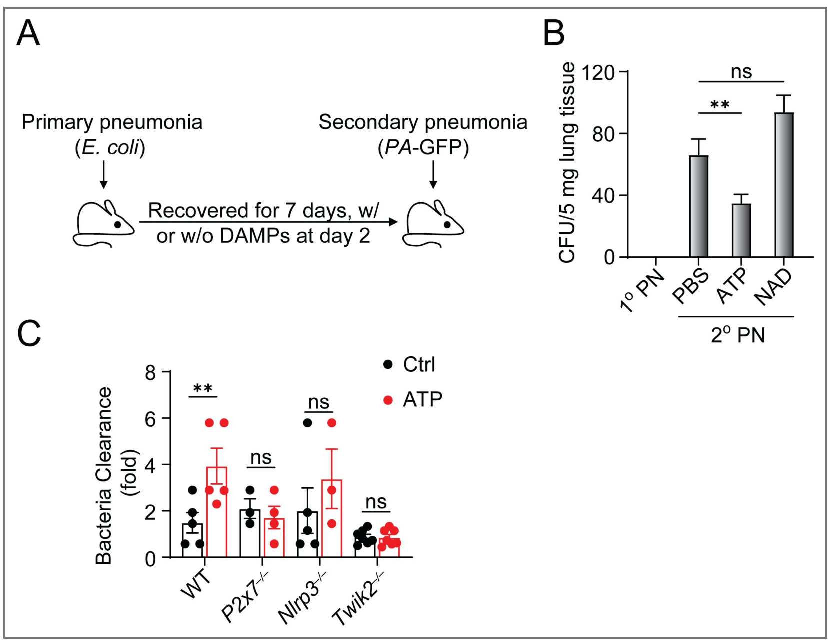


Figure 6. ATP training rescues lung immunosuppression caused by pneumonia.

A. Diagram of the double-infection model of primary pneumonia (1° PN) with 1×10^6 *E. coli* and secondary pneumonia (2° PN) with 1×10^6 GFP-*PA*. **B.** Colony-forming units (CFU) per 5mg of lung homogenate with 1° or 2° pneumonia. ∗, $p < 0.05$. **C.** Relative bacterial killing ability in lung homogenate of WT or designated KO mice trained with or without ATP in the double exposure model. Data are from (B, C) three independent experiments, mean and s.e.m. in B, C.

Discussion

Lungs are continuously exposed to environmental pathogens and require a rapid immune response to ensure host survival. Lung resident macrophages such as AMϕ are key cells of this innate defense, and induction of trained innate immunity in these cells enhances their capability. Leveraging the trained immunity of macrophages would be beneficial to promote host defense function while preventing auto-inflammatory injury, the key feature of ALI/ARDS. we discovered that ATP could train macrophages through the ATP-P2X7-TWIK2 pathway described above. Our previous data suggest that during homeostasis, TWIK2 resides in the endosomal compartment (Huang et al., 2023). In response to increased extracellular ATP, endosomal TWIK2 is translocated to the PM and remains localized in the phagosome membrane upon phagocytosis of bacteria. This raises the intriguing concept of augmenting the activity of TWIK2 that will train macrophages in multiple ways. We posit that a noteworthy element of this function is the tunability of TWIK2 association with PM and phagosomes and hence the ability of TWIK2 localization to control innate immunity for a relatively long duration (the definition of training). We hypothesize that tunability of macrophages is a central function of these cells and it requires upregulation and/or duration of ATP-mediated TWIK2 PM translocation. Therefore, controlling the magnitude and duration of TWIK2-mediated NLRP3 inflammasome activation provides potential therapeutic targets for the treatment of ALI.

Our data showed other DAMPs such as histone or NAD did not elicit trained immunity of macrophages indicating the importance of ATP signaling. Different pathogens or their products exploit distinct mechanisms to train macrophages. For example, adenoviral infection induced macrophage trained immunity requires IFN-γ produced by T-cells (Yao et al., 2018), while LPS-induced AMϕ training depends on type 1 interferon, IFN-β (Zahalka et al., 2022). LPS trained AMϕ also exhibited upregulated expression of the efferocytosis receptor MERTK and greater capacity for efferocytosis of cellular debris (Chakraborty et al., 2023). Thus, generalizations may not be possible. In addition, since viral, LPS or bacterial challenge can damage other nearby tissue and therefore generate ATP, it will be interesting to test whether ATP-TWIK2 contributes to different means of training.

There is the possibility that training and suppression are both induced upon different pathogen challenges (Verma et al., 2020; Yao et al., 2018). Both the nature of the pathogen and the pathogen dose play critical roles in determining these opposing effects (Bauer et al., 2018). For example, low-dose LPS promotes hyperactive responses while high-dose LPS prolongs inhibition in macrophages (Bauer et al., 2018; Fu et al., 2012; Zahalka et al., 2022). In addition, trained innate immunity is different from a primed immune response. In contrast to priming, trained immunity returns to the basal level following removal of the first stimulus (Divangahi et al., 2021). However, in response to secondary challenges, both gene transcription and cell functions are enhanced at much higher levels than those in the primary challenge (Divangahi et al., 2021). In our training model, the functional program of macrophages returns to basal level after first stimulation. The training model is fundamentally different from models of priming in which the stimulus is either maintained for a long period of time to induce priming or secondary stimulation is performed very quickly after the initial priming.

The discovery of trained immunity has profound implications for both immunity to infectious diseases and the development of novel therapeutic strategies. Despite the benefits, there are concerns about the potential harmful effects of a chronically trained, or partially activated, immune system (Netea et al., 2016). Thus, Understanding the mechanisms that govern the induction, persistence, and resolution of trained immunity is essential for developing strategies to harness its benefits while minimizing its risks. Our findings reveal that regulatable association of TWIK2 with the PM and phagosomes triggered by ATP mediates trained immunity in macrophages. The tunability of macrphages is a central function of these cells and it requires upregulation and/or duration of ATP-mediated TWIK2 PM translocation, which will lead to understanding new immunotherapeutic approaches.

Methods

Cell Culture and Mice

The RAW 264.7 mouse macrophage cell line was obtained from ATCC (TIB-71). RAW 264.7 cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin and were maintained at 37°C in a humidified atmosphere with 5% CO_2. For in vivo tracking of TWIK2, RAW264.7 cells were transfected with TWIK2-EGFP plasmid (pLV[Exp]-Puro-CMV >mKcnk6[NM_001033525.3]/3xGS/EGFP), designed by and purchased from VectorBuilder (VB200618-1166ypg) and cultured in puromycin selection media. Transfected cells were further selected by GFP expression. Bone marrow-derived macrophages (BMDMs) were isolated from the femurs and tibias of C57B/L6J or C57B/L6J-derived mice (8-16 weeks old, males and females harvested approximately equally) and differentiated using 0.22 µm-filtered 10% L929-conditioned media. BMDMs were cultured in RPMI 1640 with 10% FBS and 1% penicillin-streptomycin. These cells were cultured and propagated as above.

All the mice were handled in accordance with NIH guidelines for the care and use of laboratory animals and UIC animal care and use committee (ACC)'s regulations. All procedures were approved by the UIC IACUC. Mice of 8–16 weeks of age, body weight ranging from 18–25 g, were used for all experiments. P2rx7$^{-/-}$, Nlrp3$^{-/-}$ mice were purchased from Jackson Laboratory. Myeloid knockdown of TWIK2 was generated by maintaining heterozygous LysM^{cre} in a TWIK2$^{f/f}$ background. Deletion of specific genes was confirmed by PCR and western blot.

ATP-mediated training

For ATP training in cell culture, ATP salt (Thermofisher L14522.06) was dissolved at 1.0M in cell culture-grade water. This stock was diluted to a final concentration of 1mM in complete RPMI media and subsequently applied to cells. Media was changed after 24 hours, and cells were maintained for an additional 6 days until challenge or harvest. Bacterial cultures for challenge (E. coli strain DH5a and/or GFP-transformed P. aeruginosa strain PA-01) were grown in Luria Broth or Luria Broth plus 0.1 mg/ml Ampicillin, respectively, for 12-16 hours. Culture density was determined by empirically determined OD600 and bacteria (or AF594-tagged zymosan beads (Thermofisher Z23374)) were diluted to 2.5×10^7 c.f.u (or beads) per mL and co-cultured with cells in culture conditions described above. Primary challenge (*E. coli*) was co-cultured in complete RPMI 1640 media for 24 hours before washing with PBS and replenishing sterile complete media. *P. aeruginosa* and Zymosan bead endpoint challenge was co-cultured in sterile PBS for 2 hours before final assessment.

Survival Study

C57B/L6J mice were anesthetized and intranasally treated with 40 µL of cell culture-grade sterile water containing the indicated amount of dissolved ATP salt. After 6 days, mice were again anesthetized to be intranasally inoculated with GFP-expressing Pseudomonas aeruginosa (strain PA-GFP-01) at a concentration of 5×10^6 c.f.u. in a total of 40 µL cell culture-grade sterile water while in a BSL2 facility. 7 animals per group were used for the survival studies.

Bacterial Killing Assay

An in vitro bacterial killing assay was conducted using ampicillin-resistant *P. aeruginosa* (strain PA-01 GFP). Macrophages were co-cultured with bacteria at a multiplicity of infection (MOI) of 1.0. After 2 hours of incubation, samples of the supernatant were assessed to evaluate remaining bacterial load. Macrophages were washed with sterile PBS, briefly incubated with gentamicin, and lysed with cell culture grade water. Serial dilutions of both samples were plated on LB agar with 0.1mg/ml Ampicillin to quantify bacterial load (supernatant) and killing (lysate). Bacterial colonies were counted after overnight incubation at 37°C. Colony numbers were normalized to 'fold killing' by dividing the average of all negative control samples by that of each experimental group to provide ratiometric comparison. To assess in vivo bacterial clearance/bacterial load, mice were anesthetized and treated intranasally as above with ATP and, later, *P. aeruginosa*. Lungs were harvested 3 hours after inoculation, weighed, and homogenized. Homogenate was then plated on selective media to assess c.f.u. per lung mass.

Microscopy

Structured illumination microscopy (SIM) was performed using an DeltaVision OMX 3D-SIM microscope (GE Healthcare) with 3D-SIM reconstruction software, using a 60×, 1.42 NA Oil PSF Objective (Olympus PlanApo N). Wild type BMDMs were stained with ANTIBODIES and imaged. Membrane GFP intensity was measured using a Leica LSM710 BiG confocal microscope with a Pecon XL TIRF S incubation system for long term imaging of cultured cells. TWIK2-GFP expressing RAW cells were imaged with a 63× Plan-Apochromat oil objective, 1.4 NA objective once daily, and the TWIK2-GFP fluorescence intensity within 1 micron of the plasma membrane was ratiometrically compared to the average TWIK2-GFP intensity of the measured cell with ImageJ software version 1.54g.

Membrane Isolation

Plasma membranes were isolated using the Subcellular Protein Fractionation Kit for Cultured Cells (Thermo Fisher Scientific 78840) according to the manufacturer's instructions, using the maximum recommended 10×10^6 trained or untrained BMDMs per sample. Fractionation was verified as described by Thermo Scientific for the Sorvall MTX Micro-Ultracentrifuge and S55-A2 Rotor. Briefly, post-nuclear lysate of ATP-trained and untrained cells was layered on a 0-30% isotonic percoll stepwise gradient and separated via ultracentrifugation at 84000×g for 30 minutes at 4°C, the membrane-containing lysate was collected from the interface, diluted in ice-cold PBS, and further ultracentrifuged at 105,000×g for 90 minutes at 4°C to clear percoll and obtain purified membranes.

In vitro Chemical challenge and assessment

To evaluate the role of metabolism on ATP effect, metabolic inhibitors 2-Deoxy-D-glucose, NG52, and PFK158 (All from SelleckChem) were applied to cells at 1mM, 10 µM, and 15 µM respectively in incomplete media (not supplemented with FBS or antibiotics). *P. aeruginosa* was then applied to culture to assess bacterial clearance as described above. To assess TWIK2 dynamics during phagocytosis, approximately 0.1 mg of AF594-conjugated zymosan particles (Thermofisher Z23374) were added to vehicle- or ATP-treated TWIK2-GFP transfected RAW264.7 cells (approximately 2 particles per cell). Positively GFP expressing cells were imaged and assessed for 1) engulfment of particles and 2) TWIK2 localization. Frequency of engulfed particles that colocalize with TWIK2-GFP was reported.

ION Potassium Green (working concentration 5 µM in DMSO; Abcam AB142806) or equivalent volume DMSO vehicle was applied to cells concurrently with zymosan beads and assessed by confocal microscopy as described above. To assess net phagosomal K^+ influx, the baseline, $AF594^-$, cellular area green fluorescence was subtracted from that of the $AF594^+$ region, with negative values indicating higher green MFI and thus low phagosomal K^+ influx.

Fluorescence Recovery After Photobleaching (FRAP)

FRAP was performed to assess the dynamics of TWIK2-GFP on phagosomal membranes. Cells were bleached using a 488 nm laser at maximum laser power, and recovery was monitored for 2 minutes with images captured every 2 seconds, using an 63x Plan-Apochromat oil objective, 1.4 NA. Analysis was conducted using ImageJ software version 1.54g, and percent recovery was determined by ratiometrically comparing each measurement to the average of two distinct pre-bleach measurements.

Seahorse ATP Production Assay

ATP production was quantified using a Seahorse XFe 96 Analyzer (Agilent Technologies). BMDMs were differentiated and trained in 60 mm dishes, detached via Accutase 48 hours prior to assessment, live cells counted by a Typan-blue adjusted Countess II automated cell counter, and seeded in a Seahorse XFe96 cell culture microplates at 5×10^4 cells per well. Rotenone/Antimycin and Oligomycin A were loaded per manufacturer's instructions, and basal and maximal respiration rates were measured, according to the manufacturer's protocol for the XF Real Time ATP Rate Assay Kit.

NLRP3 Oligomerization Assay

Cells were washed with cold, sterile PBS and lysed in RIPA buffer (western blotting) or NP40 buffer with 1 mM DSP crosslinker (NLRP3 oligomerization). For NLRP3 oligomerization experimetns, lysates were centrifuged at 10,000 r.p.m. for 20 min to pellet the insoluble fraction and insoluble lysates were washed and diluted into NP-40 lysis buffer. Approximately equivalent amount of protein was loaded and separated on 8–12% polyacrylamide gels and transferred onto activated PVDF membranes. Membranes were blocked in 5% skim milk then incubated with primary antibodies and secondary horseradish peroxidase (HRP)-conjugated antibodies. Antibodies used were anti-Nlrp3 (Adipogen, AG-20B-0014-C100), and anti-TWIK2 (LS Bio LS-C110195-100). Actin, GapDH, or ATP1A1 were used to verify loading accuracy as indicated. Blots were developed on xray film and scanned at 300 dpi. Intensities/blots were otherwise unmodified. Band intensities were quantified using ImageJ by selecting rectangular regions of interest around each band and subtracting background signal from adjacent areas.

RNA-seq analysis

Analysis was adapted from Reference-based RNA-Seq data analysis (Galaxy Training Materials, *https://training.galaxyproject.org/training-material/topics/transcriptomics/tutorials/ref-based/tutorial.html*). In brief, RNA was isolated from ATP-treated and vehicle-treated populations of BMDMs via a Qiagen RNeasy mini kit. Concentration was normalized and raw paired-end RNA-seq reads were obtained in FASTQ format via an Illumina NovaSeqX [Uchicago genomics facility]. Sequences were uploaded to a Galaxy server and processered therein. Quality control was performed using FastQC, and adapter trimming was conducted with CutAdapt using default parameters. Trimmed reads were aligned to the reference genome using RNA STAR. Genome and annotation files were retrieved from Ensembl and indexed prior to alignment. Aligned reads were quantified using featureCounts to generate gene-level count matrices. Differential expression analysis was performed using DESeq2, with significance thresholds set at adjusted p-value < 0.05. Gene Ontology enrichment analysis was conducted using goSeq to assess biological relevance. Raw RNA sequencing data submitted to the NIH sequence read archive - PRJNA1356819.

ATAC-seq analysis

Raw reads were mapped to the mouse reference genome (mm39) using BWA MEM (Li, 2013). Read alignments were adjusted to account for TAC transposon binding: +4bp for + strand alignments, - 5bp for - strand alignments. The open chromatin enrichment track was generated by creating a bedGraph from the raw reads using bedtools genomcov (Quinlan and Hall, 2010), then coverted to bigWig using UCSC tool bedGraphToBigWig (Kent et al., 2010); tracks were normalized by the sum of alignment lengths over 1 billion. The start position track was generated by taking just the first base of the alignment for the positive strand alignments or the last base of the alignment for the negative strand alignments, then creating bedGraph and bigWig tracks as for the open chromatin; tracks were normalized to the alignment count over 1 million. Open chromatin peaks were called using Macs2 (Zhang et al., 2008) with -- nomodel set and no background provided; peaks with a score >5 were retained.

For differential open chromatin analysis, differential analysis of quantitated peaks as compared with treatment was performed using the software package edgeR on raw peak counts (McCarthy et al., 2012). Data were normalized as counts per million and an additional normalization factor was computed using the TMM algorithm. To account for different processing batches, a batch-correction factor was added to the model for the analysis. Statistical tests were performed using the "exactTest" function in edgeR. Adjusted p values (q values) were calculated using the Benjamini-Hochberg false discovery rate (FDR) correction (Benjamini and Hochberg, 1995). Significant peaks were determined based on an FDR threshold of 5% (0.05).

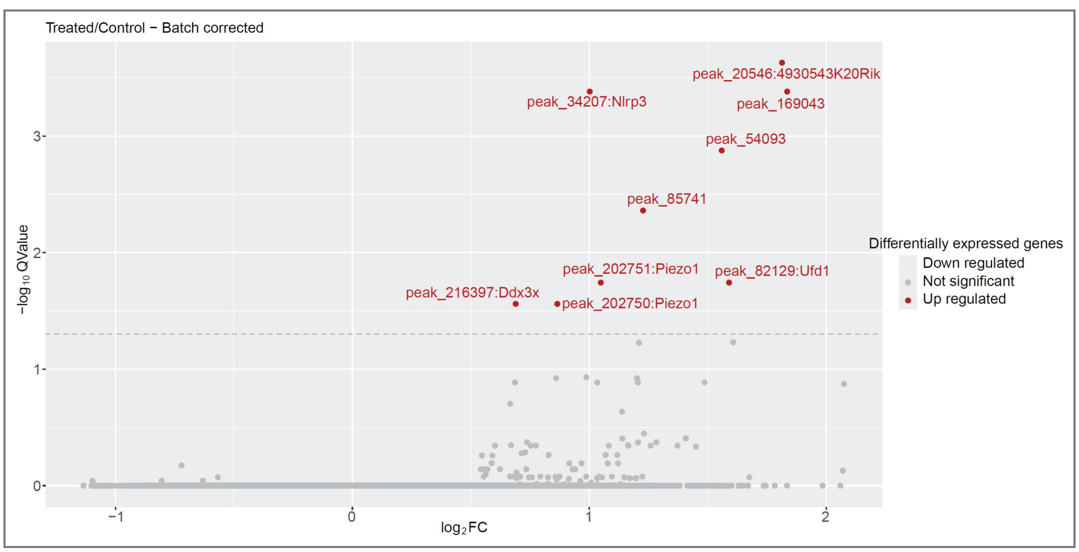
For pathway enrichment analysis, gene sets for pathway enrichment analysis were obtained from promoter-associated open chromatin, with the top 400 up- and top 400 down-regulated peaks by p value included in the analysis. A core analysis targeting upstream regulators and pathway enrichment based on this gene list was performed using Ingenuity Pathway Analysis (IPA; QIAGEN Inc, *https://digitalinsights.qiagen.com/IPA*).

For motif enrichment analysis, instances of known transcription factor motifs were searched for in called peak sequences with comparison to the JASPAR database (Khan et al., 2018) using FIMO (Bailey et al., 2009). Motif enrichment statistics were computed for the top 1000 up- and top 1000 down-regulated peaks using Fisher's Exact Test. This was repeated for all motifs, correcting for multiple testing using the FDR correction of Benjamini and Hochberg (Benjamini and Hochberg, 1995).

The ATAC seq has been submitted to the NIH SRA at SUB15765984.

Statistical analysis

All experiments were performed at least three times. Graphs were generated using GraphPad Prism. All data were expressed as mean ± standard error of the mean. Statistical comparisons were made using ordinary one-way ANOVA multiple comparisons with Prism 6 (GraphPad). Number of samples or mice per group, replication in independent experiments and statistical tests are mentioned above or in the figure legends. Significance between groups was labeled with asterisks indicating a statistically significant difference.



Supplementary Fig. 1. Volcano plot displaying $\log_2$ fold change of differentially accessible chromatin in ATP treated BMDMs relative to that of vehicle-treated BMDMs 6 days after treatment. The x-axis shows $\log_2$(fold change), and the y-axis denotes $-\log_{10}$(Q-value). Genes with both P < 0.05 and Q < 0.05 (Benjamini–Hochberg FDR) are highlighted in red and labeled with the gene containing the nearest likely promoter region. Horizontal dashed lines indicate thresholds for significance. The ATAC seq has been submitted to the NIH SRA at SUB15765984.

Data availability

RNA sequence data were deposited in NIH sequence read archive - PRJNA1356819. The ATAC sequence data were submitted to the NIH SRA at SUB15765984

Acknowledgements

We would like to thank Jason M. Wood and Mark Maienschein-Cline at Research Informatics Core, Research Resources Center, University of Illinois Chicago for help with the ATAC seq analysis. This study is supported in part by National Institute of Health grants P01HL151327 and R01DK126753 (K.K); National Natural Science Foundation of China 82130001 and Shanghai Municipal Science and Technology Major Project ZD2021CY001 (Y.S). J.T and J.C were supported by NIH T32 HL007829.

Additional information

Funding

Funder	Grant reference number	Author
HHS | NIH | NIH Office of the Director (OD)	P01HL151327	Ki-Wook Kim
HHS | NIH | NIH Office of the Director (OD)	R01DK126753	Ki-Wook Kim
National Natural Science Foundation of China	82130001	Yuanlin Song
Shanghai Municipal Science and Technology Major Project	ZD2021CY001	Yuanlin Song

Author ORCID iDs

Ki-Wook Kim: https://orcid.org/0000-0002-7207-4389
Jingsong Xu: https://orcid.org/0000-0001-8261-9509

References

Adriouch S., Haag F., Boyer O., Seman M., Koch-Nolte F. (2012) Extracellular NAD(+): a danger signal hindering regulatory T cells. *Microbes Infect* **14**:1284-1292

Allard B., Panariti A., Martin J.G. (2018) Alveolar Macrophages in the Resolution of Inflammation, Tissue Repair, and Tolerance to Infection. *Front Immunol* **9**:1777

Bailey T.L., Boden M., Buske F.A., Frith M., Grant C.E., Clementi L., Ren J., Li W.W., Noble W.S. (2009) MEME SUITE: tools for motif discovery and searching. *Nucleic Acids Res* **37**:W202-208

Bain C.C., MacDonald A.S. (2022) The impact of the lung environment on macrophage development, activation and function: diversity in the face of adversity. *Mucosal Immunol* **15**:223-234

Bauer M., Weis S., Netea M.G., Wetzker R. (2018) Remembering Pathogen Dose: Long-Term Adaptation in Innate Immunity. *Trends in immunology* **39**:438-445

Benjamini Y., Hochberg Y. (1995) Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society. Series B (Methodological)* **57**:289-300

Chakraborty S., Singh A., Wang L., Wang X., Sanborn M.A., Ye Z., Maienschein-Cline M., Mukhopadhyay A., Ganesh B.B., Malik A.B., *et al.* (2023) Trained immunity of alveolar macrophages enhances injury resolution via KLF4-MERTK-mediated efferocytosis. *J Exp Med* **220**

Coutinho-Silva R., Savio L.E.B. (2021) Purinergic signalling in host innate immune defence against intracellular pathogens. *Biochem Pharmacol* **187**:114405

Di A., Xiong S., Ye Z., Malireddi R.K.S., Kometani S., Zhong M., Mittal M., Hong Z., Kanneganti T.D., Rehman J., *et al.* (2018) The TWIK2 Potassium Efflux Channel in Macrophages Mediates NLRP3 Inflammasome-Induced Inflammation. *Immunity* **49**:56-65.

Divangahi M., Aaby P., Khader S.A., Barreiro L.B., Bekkering S., Chavakis T., van Crevel R., Curtis N., DiNardo A.R., Dominguez-Andres J., *et al.* (2021) Trained immunity, tolerance, priming and differentiation: distinct immunological processes. *Nat Immunol* **22**:2-6

Enyedi P., Czirjak G. (2010) Molecular background of leak K+ currents: two-pore domain potassium channels. *Physiol Rev* **90**:559-605

Finucane O.M., Sugrue J., Rubio-Araiz A., Guillot-Sestier M.V., Lynch M.A. (2019) The NLRP3 inflammasome modulates glycolysis by increasing PFKFB3 in an IL-1beta-dependent manner in macrophages. *Scientific reports* **9**:4034

Franchi L., Kanneganti T.D., Dubyak G.R., Nunez G. (2007) Differential requirement of P2X7 receptor and intracellular K+ for caspase-1 activation induced by intracellular and extracellular bacteria. *J Biol Chem* **282**:18810-18818

Fu Y., Glaros T., Zhu M., Wang P., Wu Z., Tyson J.J., Li L., Xing J. (2012) Network topologies and dynamics leading to endotoxin tolerance and priming in innate immune cells. *PLoS computational biology* **8**:e1002526

Grailer J.J., Canning B.A., Kalbitz M., Haggadone M.D., Dhond R.M., Andjelkovic A.V., Zetoune F.S., Ward P.A. (2014) Critical role for the NLRP3 inflammasome during acute lung injury. *J Immunol* **192**:5974-5983

He Y., Hara H., Nunez G. (2016) Mechanism and Regulation of NLRP3 Inflammasome Activation. *Trends Biochem Sci* **41**:1012-1021

Huang L.S., Anas M., Xu J., Zhou B., Toth P.T., Krishnan Y., Di A., Malik A.B. (2023) Endosomal trafficking of two-pore K(+) efflux channel TWIK2 to plasmalemma mediates NLRP3 inflammasome activation and inflammatory injury. *eLife* **12**

Jung J., Lee L.E., Kim H., Kim J.E., Jang S.H., Roh J.S., Lee B., Robinson W.H., Sohn D.H., Pyun J.C., *et al.* (2022) Extracellular histones aggravate autoimmune arthritis by lytic cell death. *Front Immunol* **13**:961197

Kent W.J., Zweig A.S., Barber G., Hinrichs A.S., Karolchik D. (2010) BigWig and BigBed: enabling browsing of large distributed datasets. *Bioinformatics* **26**:2204-2207

Khan A., Fornes O., Stigliani A., Gheorghe M., Castro-Mondragon J.A., van der Lee R., Bessy A., Cheneby J., Kulkarni S.R., Tan G., *et al.* (2018) JASPAR 2018: update of the open-access database of transcription factor binding profiles and its web framework. *Nucleic Acids Res* **46**:D1284

Li H. (2013) Aligning Sequence Reads, Clone Sequences and Assembly Contigs with BWA-MEM. *arXiv* 1303.3997

McCarthy D.J., Chen Y., Smyth G.K. (2012) Differential expression analysis of multifactor RNA-Seq experiments with respect to biological variation. *Nucleic Acids Res* **40**:4288-4297

Metzger D.W., Sun K. (2013) Immune dysfunction and bacterial coinfections following influenza. *J Immunol* **191**:2047-2052

Munoz-Planillo R., Kuffa P., Martinez-Colon G., Smith B.L., Rajendiran T.M., Nunez G. (2013) K(+) efflux is the common trigger of NLRP3 inflammasome activation by bacterial toxins and particulate matter. *Immunity* **38**:1142-1153

Netea M.G., Joosten L.A., Latz E., Mills K.H., Natoli G., Stunnenberg H.G., O'Neill L.A., Xavier R.J. (2016) Trained immunity: A program of innate immune memory in health and disease. *Science* **352**:aaf1098

Netea M.G., Joosten L.A.B. (2018) Trained Immunity and Local Innate Immune Memory in the Lung. *Cell* **175**:1463-1465

Petrilli V., Papin S., Dostert C., Mayor A., Martinon F., Tschopp J. (2007) Activation of the NALP3 inflammasome is triggered by low intracellular potassium concentration. *Cell Death Differ* **14**:1583-1589

Quinlan A.R., Hall I.M. (2010) BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* **26**:841-842

Reeves E.P., Lu H., Jacobs H.L., Messina C.G., Bolsover S., Gabella G., Potma E.O., Warley A., Roes J., Segal A.W. (2002) Killing activity of neutrophils is mediated through activation of proteases by K+ flux. *Nature* **416**:291-297

Sefik E., Qu R., Junqueira C., Kaffe E., Mirza H., Zhao J., Brewer J.R., Han A., Steach H.R., Israelow B., *et al.* (2022) Inflammasome activation in infected macrophages drives COVID-19 pathology. *Nature*

Sevransky J.E., Martin G.S., Shanholtz C., Mendez-Tellez P.A., Pronovost P., Brower R., Needham D.M. (2009) Mortality in sepsis versus non-sepsis induced acute lung injury. *Crit Care* **13**:R150

Swanson K.V., Deng M., Ting J.P. (2019) The NLRP3 inflammasome: molecular activation and regulation to therapeutics. *Nat Rev Immunol* **19**:477-489

Tapia-Abellan A., Angosto-Bazarra D., Alarcon-Vila C., Banos M.C., Hafner-Bratkovic I., Oliva B., Pelegrin P. (2021) Sensing low intracellular potassium by NLRP3 results in a stable open structure that promotes inflammasome activation. *Science advances* **7**:eabf4468

Verma A.K., Bansal S., Bauer C., Muralidharan A., Sun K. (2020) Influenza Infection Induces Alveolar Macrophage Dysfunction and Thereby Enables Noninvasive Streptococcus pneumoniae to Cause Deadly Pneumonia. *J Immunol* **205**:1601-1607

Vieira O.V., Botelho R.J., Grinstein S. (2002) Phagosome maturation: aging gracefully. *Biochem J* **366**:689-704

Yao Y., Jeyanathan M., Haddadi S., Barra N.G., Vaseghi-Shanjani M., Damjanovic D., Lai R., Afkhami S., Chen Y., Dvorkin-Gheva A., *et al.* (2018) Induction of Autonomous Memory Alveolar Macrophages Requires T Cell Help and Is Critical to Trained Immunity. *Cell* **175**:1634-1650.

Zahalka S., Starkl P., Watzenboeck M.L., Farhat A., Radhouani M., Deckert F., Hladik A., Lakovits K., Oberndorfer F., Lassnig C., *et al.* (2022) Trained immunity of alveolar macrophages requires metabolic rewiring and type 1 interferon signaling. *Mucosal Immunol* **15**:896-907

Zhang Y., Liu T., Meyer C.A., Eeckhoute J., Johnson D.S., Bernstein B.E., Nusbaum C., Myers R.M., Brown M., Li W., *et al.* (2008) Model-based analysis of ChIP-Seq (MACS). *Genome biology* **9**:R137

Zhu G., Yu H., Peng T., Yang K., Xu X., Gu W. (2024) Glycolytic enzyme PGK1 promotes M1 macrophage polarization and induces pyroptosis of acute lung injury via regulation of NLRP3. *Respir Res* **25**:291

Peer reviews

Reviewer #1 (Public review):

Summary:

Alveolar macrophages (AMs) are key sentinel cells in the lungs, representing the first line of defense against infections. There is growing interest within the scientific community in the metabolic and epigenetic reprogramming of innate immune cells following an initial stress, which alters their response upon exposure to a heterologous challenge. In this study, the authors show that exposure to extracellular ATP can shape AM functions by activating the P2X7 receptor. This activation triggers the relocation of the potassium channel TWIK2 to the cell surface, placing macrophages in a heightened state of responsiveness. This leads to the activation of the NLRP3 inflammasome and, upon bacterial internalization, to the translocation of TWIK2 to the phagosomal membrane, enhancing bacterial killing through pH modulation. Through these findings, the authors propose a mechanism by which ATP acts as a danger signal to boost the antimicrobial capacity of AMs.

Strengths:

This is a fundamental study in a field of great interest to the scientific community. A growing body of evidence has highlighted the importance of metabolic and epigenetic reprogramming in innate immune cells, which can have long-term effects on their responses to various inflammatory contexts. Exploring the role of ATP in this process represents an important and timely question in basic research. The study combines both in vitro and in vivo investigations and proposes a mechanistic hypothesis to explain the observed phenotype.

Weaknesses:

The authors have revised the manuscript to address the comments raised during the first round of review. However, several figures, figure legends, and methodological sections still require additional adjustments and clarification.

The interpretation of CFU from lysates as 'killing' is unclear; lysate CFUs typically reflect intracellular surviving bacteria and are confounded by differences in uptake. Please include an uptake control (early time point) or time-course to distinguish phagocytosis from intracellular killing. Also, clarify how bacterial burden was calculated (supernatant vs cell-associated vs total). Supernatant alone may not capture adherent bacteria. The normalization as 'fold killing' (mean negative control / sample) is non-standard; please report absolute CFU (log scale) and specify the exact definition of killing/survival.

The Methods section is largely incomplete and requires substantial revision. For instance, the authors report quantification of cytokine concentrations, yet no information is provided regarding how these measurements were performed. It is unclear whether cytokines were measured in BALF by ELISA, or assessed at the mRNA level by qPCR from total lung lysates, or by another method. This information must be clearly specified. In addition, the rationale for selecting the measured cytokines should be justified. While the choice of IL-1β and IL-6 is relatively straightforward, the focus on IL-18 requires explicit justification.

Similarly, the methodology used to quantify immune cell populations presented in Figure 2 is not described. It is not stated how immune cells were isolated and identified (e.g. flow cytometry from lung tissue). No information is provided regarding tissue digestion, cell isolation procedures, or gating strategy (presumably by flow cytometry). These details are essential and should be included, together with the corresponding gating strategy and absolute cell numbers.

Moreover, immune cell quantification would be expected in the context of the challenge experiment as well. Reporting unchanged percentages of lung immune cells following ATP exposure does not support the conclusion of a training effect, particularly one that is specific to alveolar macrophages (AMs). In addition, AMs are not considered recruited immune cells; this should be corrected in the figure legend and throughout the manuscript where applicable.

There are inconsistencies throughout the manuscript. For example, the authors report n = 5 for the survival curves in the figure legend, whereas n = 7 is stated in the Methods section. This discrepancy is unclear and should be clarified.

Supplementary Fig. 1 contains major conceptual errors. The volcano plot represents ATAC-seq peaks (differentially accessible chromatin regions), yet the figure, legend, and color scale repeatedly refer to 'genes' and 'differentially expressed genes'. This conflates chromatin accessibility with gene expression and is misleading. Peaks are secondarily annotated to nearby genes, which should be clearly described as an annotation step rather than the unit of analysis. The figure should be revised to explicitly present peak-level statistics (DARs), with gene names shown only as optional annotations. Additionally, the use of simultaneous P <

0.05 and Q < 0.05 thresholds is non-standard, and the absence of down-regulated regions in the plot requires explanation.

In Figure 7, trained WT and Nlrp3^{-}/$^{-}$ mice display similar levels of bacterial clearance. How should this result be interpreted?

Overall, while the study addresses an interesting biological question, the manuscript would benefit from substantial revision prior to publication. In particular, clarifications and improvements regarding the methodology, data presentation, and interpretation are required to strengthen the rigor and reproducibility of the conclusions.

https://doi.org/10.7554/eLife.107536.2.sa2

Reviewer #2 (Public review):

Summary:

In this manuscript, Thompson et al. investigate the impact of prior ATP exposure on later macrophage functions as a mechanism of immune training. They describe that ATP training enhances bactericidal functions which they connect to the P2x7 ATP receptor, Nlrp3 inflammasome activation, and TWIK2 K+ movement at the cell surface and subsequently at phagosomes during bacterial engulfment. This is an incremental addition to existing literature, which has previously explored how ATP alters TWIK2 and K+, and linked it to Nlrp3 activation. The novelty here is in discovering the persistence of TWIK2 change and exploring the impact this biology may have on bacterial clearance. Additional experiments could strengthen their hypothesis that the in vivo protective effect of ATP-training on bacterial clearance is mediated by alveolar macrophages.

Strengths:

The authors demonstrate three novel findings: 1) prolonged persistence of TWIK2 at the macrophage plasma membrane following ATP that can translocate to the phagosome during particle engulfment, 2) a persistent impact of ATP exposure on remodeling chromatin around nlrp3, and 3) administering mice intra-nasal ATP to 'train' lungs protects mice from otherwise fatal bacterial infection.

Weaknesses:

(1) Some methods remain unclear including the timing and method by which lung cellularity was assessed in Figure 2. It is also difficult to understand how many mice were used in experiments 1, 2 and 6 and thus how rigorous the design was. A specific number is only provided for 1D and the number of mice stated in legend and methods do not match.

(2) The study design is not entirely ideal for the authors' in vivo question. Overall, the discussion would benefit from a clear summary of study caveats, which are primarily that that 1) in vitro studies attributing ATP training-mediated bacterial killing to persistent TWIK2 relocation, K+ influx, a glycolytic metabolic shift , and epigenetic nlrp3 reprogramming were performed in BMDM or RAW cells and not primary AMs, 2) data does not eliminate the possibility that non-AM immune or non-immune cells in the lung are "trained" and responsible for ATP-mediated protection in vivo; flow data examined total lung digest which may obscure important changes in alveolar recruitment, and 3) in vivo work shows data on acute bacterial clearance but does not explore potential risks that "training" for a more responsive inflammasome may have for the severity of lung injury during infection.

(3) The is some lack of transparency on data and rigor of methods. Clear data is not provided regarding the RNA-sequencing results. Specific identities of DEGs is not provided, only one high-level pathway enrichment figure. It would also be ideal if controls were included for

subcellular fractionating to confirm pure fractions and for dye microscopy to show negative background.

(4) In results describing 5A, the text states that "ATP-induced macrophage training effects, as measured by augmented bactericidal activity, were diminished in macrophages treated with protease inhibitors". However, these data are not identified significant in the figure; protease dependence can be described as a trend that supports the authors' hypothesis but should not be stated as significant data in text.

In summary, this work contains some useful data showing how ATP can train macrophages via TWIK2/Nlrp3. Revisions have significantly improved methods reporting, added some data to strengthen the conclusions, and toned down on overstatements to bring conclusions more in line with data presented. The title still overstates what the authors have actually tested, since no macrophage-specific targeting in vivo (no conditional gene deletion, macrophage depletion etc) was performed in infection studies. However, in vitro data provide clear evidence that macrophages can be trained by ATP, and through caveats remain, it is plausible that macrophage training is a key mechanism for the protection observed here in the lung.

https://doi.org/10.7554/eLife.107536.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *(1) First, the concept of training or trained immunity refers to long-term epigenetic reprogramming in innate immune cells, resulting in a modified response upon exposure to a heterologous challenge. The investigations presented demonstrate phenotypic alterations in AMs seven days after ATP exposure; however, they do not assess whether persistent epigenetic remodeling occurs with lasting functional consequences. Therefore, a more cautious and semantically precise interpretation of the findings would be appropriate.*

In response, we have performed epigenetic analysis (ATAC seq analysis) as requested (Supp Fig. 1).

> *(2) Furthermore, the in vivo data should be strengthened by additional analyses to support the authors' conclusions. The authors claim that susceptibility to Pseudomonas aeruginosa infection differs depending on the ATP-induced training effect. Statistical analyses should be provided for the survival curves, as well as additional weight curves or clinical assessments. Moreover, it would be appropriate to complement this clinical characterization with additional measurements, such as immune cell infiltration analysis (by flow cytometry), and quantification of pro-inflammatory cytokines in bronchoalveolar lavage fluid and/or lung homogenates.*

We have added the statistical analyses provided for the survival curves (new Fig. 1D), immune cell infiltration analysis, and quantification of pro-inflammatory cytokines in the lung (new Figs. 1, 2).

> *(3) Moreover, the authors attribute the differences in resistance to P. aeruginosa infection to the ATP-induced training effect on AMs, based on a correlation between in vivo survival curves and differences in bacterial killing capacity measured in vitro. These are correlative findings that do not establish a causal role for AMs in the in vivo phenotype. ATP-mediated effects on other (i.e., non-AM) cell populations are omitted, and*

> *the possibility that other cells could be affected should be, at least, discussed. Adoptive transfer experiments using AMs would be a suitable approach to directly address this question.*

We have performed additional experiments and found that the numbers of lung macrophages were not significantly altered before and after ATP training (new Fig. 2), indicating the training effects are focused on lung resident macrophages.

> ***Reviewer #2 (Public review):***
>
> *(1) Missing details from methods/reported data: Substantial sections of key methods have not been disclosed (including anything about animal infection models, RNA-sequencing, and western blotting), and the statistical methods, as written, only address two-way comparisons, which would mean analysis was improperly performed. In addition, there is a general lack of transparency - the methods state that only representative data is included in the manuscript, and individual data points are not shown for assays.*

We have revised the methods and statistical analysis.

> *(2) Poor experimental design including missing controls: Particularly problematic are the Seahorse assay data (requires normalization to cell numbers to interpret this bulk assay - differences in cell growth/loss between conditions would confound data interpretation) and bacterial killing assays (as written, this method would be heavily biased by bacterial initial binding/phagocytosis which would confound assessment of killing). Controls need to be included for subcellular fractionating to confirm pure fractions and for dye microscopy to show a negative background. Conclusions from these assays may be incorrect, and in some cases, the whole experiment may be uninterpretable.*

Seahorse assay methodology was updated to confirm the order of cell counting, time at seeding and cell counts. Methods were also updated to address the distinction between bacterial killing (Fig. 1B) and overall decrease in bacterial load.

> *(3) The conclusions overstate what was tested in the experiments: Conceptually, there are multiple places where the authors draw conclusions or frame arguments in ways that do not match the experiments used. Particularly:*
>
> *(a) The authors discuss their findings in the context of importance for AM biology during respiratory infection but in vitro work uses cells that are well-established to be poor mimics of resident AMs (BMDM, RAW), particularly in terms of glycolytic metabolism.*

We have adjusted the text to reflect that the metabolic assay was performed on BMDMs. AMs are fragile for certain manipulations in vitro. We expect that the metabolic change is similar across several macrophage systems as well as the bacterial load reduction.

> *(b) In vivo work does not address whether immune cell recruitment is triggered during training.*

We have performed immune cell infiltration analysis (new Fig. 2).

> *(c) Figure 3 is used to draw conclusions about K+ in response to bacterial engulfment, but actually assesses fungal zymosan particles.*

We have corrected this in the manuscript.

> *(d) Figure 5 is framed in bacterial susceptibility post-viral infection, but the model used is bacterial post-bacterial.*

We have corrected this in the manuscript.

> *(e) In their discussion, the authors propose to have shown TWIK2-mediated inflammasome activation. They link these separately to ATP, but their studies do not test if loss of TWIK2 prevents inflammasome activation in response to ATP (Figure 4E does not use TWIK2 KO).*

We have now added the TWIK2 KO results (new Fig. 5E).

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *As noted in the public review, it would be advisable to further characterize the in vivo phenotype in order to strengthen the conclusions. Specifically, it would be useful to quantify the bacterial load in the bronchoalveolar lavage fluid and lung homogenates, as well as to measure cytokine levels both in the respiratory compartment and systemically. Additionally, a broader characterization of the immune response in the presence or absence of ATP-induced training would be valuable. In the absence of direct evidence demonstrating that trained AMs mediate the observed phenotype, the authors should adopt a more cautious interpretation of their results. Moreover, careful attention to semantic accuracy is recommended. The concept of trained immunity refers specifically to long-term epigenetic reprogramming that leads to an altered response of target cells upon a secondary challenge, distant from the initial stress. The data presented do not fully demonstrate this phenomenon, and the interpretations should remain aligned with the evidence provided.*

Bacterial load has been quantified (see more details in the Methods part). And we also measured immune cell infiltration, quantification of pro-inflammatory cytokines in the lung (new Figs. 1, 2), and epigenetic evaluation of vehicle- and ATP-treated cells (Supp. Fig. 1).

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) It cannot be overstated how lacking the methods are. This includes no discussion of IACUC approval for animal procedures, which must be included as part of research ethics. It also needs to be made clear where raw data is being archived. This notably includes an accession for deposited RNA-sequencing data, although unmanipulated microscopy and western blot images should also be shown. Methods should discuss any pre-processing that occurred with images.*

We have revised the methods in the manuscript.

> *(2) Per statistics, in addition to generally providing more detail and adjusting analyses if they have not been correctly performed, please disclose if SD or SEM is shown. Reporting aggregate data versus representative data would provide more rigor. Perhaps replicate experiments could be included in the supplemental if they cannot, for some reason, be aggregated. Detailed statistical methods for RNA-seq analysis also need to be included.*

More details have been provided in the methods section.

> *(3) It is unclear whether bacterial killing assays were correctly designed and can be interpreted. What does cells collected mean? If the assay was focused on intracellular macrophage bacterial load, it is critical to assess and report phagocytosis since different input loads would confound the assessment of killing. A rigorous wash or an antibiotic to eliminate extracellular bacteria should also have been performed and be described in this case. If the total bacterial burden was assessed, that would use cells+media and also needs to be clear and described. With the information provided, it is unclear whether the assays performed are sufficiently rigorous to assess bacterial killing. In addition, Figure 1B reports using an MOI of 50-100, but all data is compiled in one graph - data from*

> *different levels of infection should be separated. Figure 5A shows a model with E.coli followed by PA, but that does not appear to be how the assay was structured in B or C. This also does not match how the experiment is written in the results section, which references S. aureus. It is unclear what tissue (or cells) were assessed in Figure 5. Whole lung? BAL? As written, no data provided regarding bacterial killing is of sufficient quality to be considered valid.*

We have re-written the bacterial killing assay in the manuscript. The methodology was corrected to distinguish bacterial killing vs load decrease and generally accurate methodology.

> *(4) The in vitro data provide reasonable evidence that BMDM/RAW macrophage training can occur in response to ATP exposure. However, it is unclear whether training is an important mechanism for resident AM in vivo, or whether, in vivo, a broader inflammatory response is generated, recruiting additional immune cells that persist and change infection susceptibility. The authors argue for resident AM immune training, but do not provide sufficient evidence to counter the latter possibility (resident AM are never themselves directly assessed, and the presence of other immune cells in vivo is not excluded). See Iliakis et al 2023 (PMID 37640788) for discussion of how this issue continues to drive uncertainty in the field. For this study, at least providing flow cytometry data quantifying myeloid and lymphoid immune populations in BALF before and after various treatments would help address this caveat. Without knowing this, it also confounds the interpretation of Figure 1B; if BAL is not pure AM after training, perhaps 1B could be repeated with ex vivo training or resident AM could be purified?*

We have performed immune cell infiltration analysis in the lung (both to BALF and in-tissue, new Fig. 2).

> *(5) Figure 3A appears to show that fewer than 50% of cells express GFP. Is it expected that only a fraction of RAW cells express TWIK2-GFP? How was this addressed in the analyses for Figure 3? Were cells not appearing to express any significant GFP, included in phagosomal-negative or excluded from analysis? Please include in the methods.*

The RAW cells were transfected with TWIK2-GFP and variable GFP expression was expected. These cells were expressing a non-integrated transgene, which has been added to the methods as well as the consideration of cells for the analysis. Cells without visible GFP expression were excluded.

> *(6) Why are many data points in Figure 3D negative? This suggests that settings were not optimized for microscopy - perhaps there is a very high background signal and the ION stain is barely above it. This is concerning for the quality of data. Further, is it expected that only some cells are positive for ION K+? The images shown clearly differentiate phagosomal K with ATP versus the absence of K without, but it is surprising that some cells appear not to contain any ION K+ signal (not completely clear given lack of brightfield or other cell staining) - this may again point to issues with imaging settings that confound data interpretation. This analysis should be carefully assessed.*

This has been updated in the methodology. In old Fig. 3D (new Fig. 4D), the presented data is the net intensity of the phagosome, subtracting the average cytoplasmic MFI from that of the area corresponding to an engulfed zymosan-af594 bead. Thus, a negative value has higher cytoplasmic IonK signal than that of the phagosome.

> *(7) The Discussion states that it will be interesting to test whether ATP-TWIK2 is a common mechanism of training and specifically references LPS as an ATP-generating signal. However, Figure 2D data show that LPS induces only transient TWIK2 translocation; the authors have data suggesting that, in the context of LPS, TWIK2*

> *'training' will not be engaged. This line of discussion shows incomplete consideration of the data.*

We have further limited this language in the text such that this may require differential sensitivity/damage sustained by macrophages as compared to that of epi/endothelial cells in response to bacterial endotoxin.

> *(8) For RNA-sequencing, plots of the actual genes changed for the mitochondrial pathways of interest would be helpful information for readers, as would a heat map showing sample purity between groups for macrophage markers versus possible contaminant cells, which can also be generated from precursors in BMDM cultures. In general, information in Methods regarding how the analyses in Figure 4B were run is necessary, per cutoffs used to determine DEGs, number of samples in each group, sex of samples used, etc. Greater transparency of data would be appreciated, so plots that show variation between replicates, such as heat maps, would be ideal. Supplemental tables would also be nice.*

We have added to the methodology of the RNA sequencing analysis

> *(9) The use of alternate DAMPs is a positive addition to the experimental design, but no data is given regarding the concentrations used. Ideally, positive controls showing histones/NAD are used at acutely activating concentrations could be included but at least references supporting the doses chosen or information about how doses were selected should be given. It is easy to find substantial literature on histones as a DAMP, but it was unclear why/how NAD was selected.*

We have added these concentrations and corresponding references.

> *(10) The E.coli CFU reported in Figure 5B are extraordinarily low. In addition, CFU are generally shown on a log scale, but this appears to be linear. Please confirm that these data are correct. Perhaps improved methods might explain why? Is the second hit a low dose?*

These have been corrected in the new Fig. 6B.

> *(11) Given that loss of either TWIK2 or Nlrp3 ablates bacterial protection, a link should be tested - experiments should test whether loss of TWIK2 prevents inflammasome activation in response to ATP (TWIK2 KO in 4E) and if loss of Nlrp3 changes TWIK2 translocation (Nlrp3 KO in at least some experiments of Figures 2/3).*

We have now added the TWIK KO results (new Fig. 5E).

> *(12) One of the most striking data pieces is Figure 1D. It would, therefore, strengthen the paper to repeat those experiments (even just with the high-dose ATP) using TEIK2/P2X7/NLRP3 KO mice and really show the importance of these pathways in vivo. This is conceptually Figure 5, but the survival data of Figure 1 is far more convincing than the relatively weak bacterial load data of Figure 5.*

Unfortunately, our previous laboratory has been closed and we have trouble acquiring enough mice for additional survival data during the transition period. However, the bacterial load data has been adjusted to the same bacterial counts per 5 mg lung tissue instead of per individual sampling, giving a more contextual interpretation of the data.

https://doi.org/10.7554/eLife.107536.2.sa0